Completeness of L^p and pointwise convergence

REAL VARIABLES I

Fall 2026

We established that the space $L^p(X,\mu)$ is a normed vector space for exponents $p \in [1,+\infty)$ and a quasinormed vector space for $p \in (0,1)$. A classical limitation of function spaces such as $C^0([a,b])$ or that of Riemann integrable functions, equipped with the L^1 metric is the failure of metric completeness. For example, Cauchy sequences of continuous functions frequently fail to converge to a continuous limit (Note 1). In a metric or normed space, completeness ensures that every sequence of approximations whose mutual differences shrink to zero converges to an element belonging to the space itself. The primary result of these notes is that $L^p(X,\mu)$ is complete for every exponent $p \in (0,+\infty)$. Consequently, $L^p(X,\mu)$ is a Banach space for every $p \in [1,+\infty)$..

To establish completeness, it is advantageous to formulate convergence in terms of the p-th power functional $\|f\|_p^p = \int_X |f|^p \, \mathrm{d}\mu$ rather than the norm $\|f\|_p$. Because $\|f\|_p^p$ is expressed directly as an integral of the non-negative pointwise quantity $|f|^p$, convergence can be analyzed using the monotone convergence theorem. The proof proceeds in three stages. First, we establish a numerical quasi-triangle inequality $(a+b)^p \le K_p(a^p+b^p)$ with constant $K_p \coloneqq \max\{1, 2^{p-1}\}$ and iterate it to bound p-th powers of finite sums. Second, we prove that every Cauchy sequence in $L^p(X,\mu)$ possesses a subsequence with geometrically decaying consecutive differences, and that convergence of such a subsequence implies convergence of the full Cauchy sequence. Third, for a rapidly Cauchy sequence, we construct the limit pointwise via an absolutely convergent telescoping series and verify that it provides the L^p limit.

The construction yields a limit function that is simultaneously an L^p limit and an almost-everywhere pointwise limit. In the second half of these notes, we systematically investigate the relationship between convergence in L^p and pointwise convergence almost everywhere. We prove that rapid L^p convergence with summable errors implies almost-everywhere convergence, and we deduce that every L^p-convergent sequence admits an almost-everywhere convergent subsequence. Conversely, we demonstrate through explicit counterexamples (including the typewriter sequence) that neither mode of convergence implies the other in general. Finally, we establish the dominated convergence theorem in L^p, which restores L^p convergence from almost-everywhere convergence under an integrable domination hypothesis.

After studying these notes, you will be able to do the following.

- State and prove the quasi-triangle inequality for the power functional $\|\cdot\|_p^p$ with constant $K_p = \max\{1, 2^{p-1}\}$, and iterate it to bound p-th powers of finite sums.
- Reduce the completeness of $L^p(X,\mu)$ to the convergence of rapidly Cauchy sequences with geometrically decaying consecutive differences.
- Prove the Riesz–Fischer theorem: $L^p(X,\mu)$ is complete for every exponent $p \in (0,+\infty)$, so that $L^p(X,\mu)$ is a Banach space for $p \in [1,+\infty)$.
- Prove that rapid L^p convergence with summable errors implies almost-everywhere convergence, and deduce that every L^p-convergent sequence admits an almost-everywhere convergent subsequence.
- Construct the typewriter sequence to demonstrate that a sequence can converge in L^p while diverging at every point.
- Construct sequences showing that almost-everywhere convergence does not imply L^p convergence, both on spaces of infinite measure (escape of mass to infinity) and on spaces of finite measure (concentration of mass).

- Apply the dominated convergence theorem in L^p to deduce L^p convergence from almost-everywhere convergence under a domination condition.

Throughout these notes, $(X, \mathcal{A}, \mu)$ denotes a measure space, $p \in (0, +\infty)$ is a fixed exponent, and functions are measurable functions $f\colon X \to \mathbb{R}$. On subsets of $\mathbb{R}$, the symbol $\mathcal{L}$ denotes Lebesgue measure.

12.1 Cauchy sequences in L^p

On the space $L^p(X, \mu)$, metric convergence is defined with respect to the metric $d_p(f, g) \coloneqq \|f - g\|_p$ for $p \in [1, +\infty)$ and with respect to $d_p(f, g) \coloneqq \|f - g\|_p^p$ for $p \in (0, 1)$ (Note 24). Because the function $t \mapsto t^p$ is a strictly increasing homeomorphism of $[0, +\infty)$ onto $[0, +\infty)$, a quantity $\|h\|_p$ tends to 0 if and only if $\|h\|_p^p$ tends to 0. We may therefore formulate both convergence and the Cauchy property directly in terms of the integral quantity $\|h\|_p^p = \int_X |h|^p \, \mathrm{d}\mu$, providing a unified framework for all exponents $p \in (0, +\infty)$.

Definition 12.1 (Convergence and Cauchy sequences in L^p). A sequence $(f_n)_{n\in\mathbb{N}}$ in $L^p(X, \mu)$ **converges in L^p** to $f \in L^p(X, \mu)$ if

$$\lim_{n\to\infty} \|f_n - f\|_p^p = \lim_{n\to\infty} \int_X |f_n - f|^p \, \mathrm{d}\mu = 0.$$

The sequence $(f_n)_{n\in\mathbb{N}}$ is a **Cauchy sequence in L^p** if for every $\varepsilon > 0$ there exists an index $N \in \mathbb{N}$ such that

$$\|f_n - f_m\|_p^p < \varepsilon \qquad \text{for all } n, m \geq N.$$

The space $L^p(X, \mu)$ is **complete** if every Cauchy sequence in $L^p(X, \mu)$ converges in L^p to an element of $L^p(X, \mu)$.

For $p \in [1, +\infty)$, Definition 12.1 coincides with completeness of the normed space $(L^p(X, \mu), \|\cdot\|_p)$. For $p \in (0, 1)$, it coincides with completeness of the metric space $(L^p(X, \mu), d_p)$ equipped with the metric $d_p(f, g) = \|f - g\|_p^p$.

12.1.1 The quasi-triangle inequality for the power functional

The analytical estimates in this unit rely on an inequality satisfied by the p-th power functional $\|\cdot\|_p^p$. We define the exponent-dependent constant

$$K_p \coloneqq \max\{1, 2^{p-1}\} = \begin{cases} 1, & 0 < p \leq 1, \\ 2^{p-1}, & p > 1. \end{cases} \tag{12.1}$$

Proposition 12.2 (Quasi-triangle inequality for $\|\cdot\|_p^p$). For all real numbers $a, b \geq 0$,

$$(a + b)^p \leq K_p(a^p + b^p). \tag{12.2}$$

Consequently, for all $f, g \in L^p(X, \mu)$,

$$\|f + g\|_p^p \leq K_p(\|f\|_p^p + \|g\|_p^p). \tag{12.3}$$

Proof of Proposition 12.2. Step 1: The numerical inequality.
We verify (12.2) according to whether $p \leq 1$ or $p > 1$. If $a + b = 0$, then $a = b = 0$ and both sides of (12.2) vanish. Suppose that $a + b > 0$. For $p \in (0, 1]$, the ratios $u \coloneqq a/(a+b)$ and $v \coloneqq b/(a+b)$ belong to

$[0,1]$ and satisfy $u+v=1$. Because $p \le 1$, the function $t \mapsto t^p$ satisfies $t^p \ge t$ for all $t \in [0,1]$. Therefore

$$\left(\frac{a}{a+b}\right)^p + \left(\frac{b}{a+b}\right)^p \ge \frac{a}{a+b} + \frac{b}{a+b} = 1.$$

Multiplying both sides by $(a+b)^p$ yields $(a+b)^p \le a^p + b^p = K_p(a^p+b^p)$ since $K_p = 1$ for $p \in (0,1]$. For $p > 1$, the second derivative $\frac{\mathrm{d}^2}{\mathrm{d}t^2} t^p = p(p-1)t^{p-2}$ is positive on $(0,+\infty)$, so $t \mapsto t^p$ is convex on $[0,+\infty)$. Applying convexity with weight $1/2$ gives

$$\left(\frac{a+b}{2}\right)^p \le \frac{1}{2}(a^p + b^p).$$

Multiplying both sides by 2^p yields $(a+b)^p \le 2^{p-1}(a^p+b^p) = K_p(a^p+b^p)$, since $K_p = 2^{p-1}$ for $p > 1$. This proves (12.2) for all $a, b \ge 0$.

Step 2: Integration over the measure space.
For $f, g \in L^p(X,\mu)$, the pointwise triangle inequality in $\mathbb{R}$ gives $|f(x)+g(x)| \le |f(x)| + |g(x)|$ at every point $x \in X$. Because $t \mapsto t^p$ is non-decreasing on $[0,+\infty)$, applying (12.2) with $a = |f(x)|$ and $b = |g(x)|$ yields

$$|f(x)+g(x)|^p \le \big(|f(x)| + |g(x)|\big)^p \le K_p(|f(x)|^p + |g(x)|^p).$$

Integrating this pointwise inequality over X with respect to μ yields (12.3). □

For $p \in (0,1]$, the constant is $K_p = 1$, so (12.3) is the subadditivity of the metric $d_p(f,g) = \|f-g\|_p^p$ established in Note 24. For $p > 1$, Minkowski's inequality gives the sharper bound $\|f+g\|_p \le \|f\|_p + \|g\|_p$, but the weaker bound (12.3) suffices for every completeness argument below and provides a unified proof across all exponents.

As an immediate consequence of (12.3), limits in $L^p(X,\mu)$ are unique. Indeed, if $f_n \to f$ and $f_n \to g$ in $L^p(X,\mu)$, then

$$\|f-g\|_p^p \le K_p(\|f-f_n\|_p^p + \|f_n-g\|_p^p) \longrightarrow 0 \qquad \text{as } n \to \infty.$$

Hence $\|f-g\|_p = 0$, which implies $f = g$ in $L^p(X,\mu)$ by the separation property of the L^p metric (Definition 24.2 / Note 24).

The completeness proof requires applying (12.2) to finite sums with an arbitrary number of terms. Iterating the two-term inequality introduces powers of the constant K_p, as recorded in the next lemma.

Lemma 12.3 (Iterated quasi-triangle inequality). For every integer $N \in \mathbb{N}$ and all real numbers $a_1, \dots, a_N \ge 0$,

$$\left(\sum_{n=1}^{N} a_n\right)^p \le \sum_{n=1}^{N} K_p^n \, a_n^p. \tag{12.4}$$

Proof of Lemma 12.3. We argue by mathematical induction on N. For $N = 1$, the inequality states that $a_1^p \le K_p a_1^p$, which holds because $K_p \ge 1$. Suppose that (12.4) holds for all sums of $N-1$ non-negative terms, where $N \ge 2$. Applying (12.2) with $a = a_1$ and $b = \sum_{n=2}^{N} a_n$, and then applying the induction hypothesis to the $N-1$ numbers $a_2, \dots, a_N$, yields

$$\left(\sum_{n=1}^{N} a_n\right)^p \le K_p a_1^p + K_p \left(\sum_{n=2}^{N} a_n\right)^p \le K_p a_1^p + K_p \sum_{n=2}^{N} K_p^{n-1} a_n^p = \sum_{n=1}^{N} K_p^n \, a_n^p.$$

This completes the induction and establishes (12.4). □

For $p \in (0,1]$, the constant is $K_p = 1$, so all weights K_p^n in (12.4) equal 1. For $p > 1$, the weights $K_p^n = 2^{n(p-1)}$ grow geometrically with n. Consequently, bounding a series $\sum_n a_n$ through (12.4) requires that the terms a_n^p decay faster than K_p^{-n}, a condition that guides the reduction lemma in the next subsection.

12.1.2 Reduction to rapidly Cauchy sequences

In a general metric space, a Cauchy sequence $(f_n)_{n\in\mathbb{N}}$ satisfies the condition that $d(f_n, f_m) \to 0$ as $n, m \to \infty$, but this condition does not specify a rate at which the differences shrink. To construct a candidate limit function as the sum of a telescoping series $f_1 + \sum_{n=1}^{\infty}(f_{n+1} - f_n)$, we require consecutive differences that decay at a prescribed geometric rate. The next lemma demonstrates that it is sufficient to establish convergence for sequences with geometric decay: every Cauchy sequence admits a subsequence with geometrically decaying differences, and whenever a subsequence of a Cauchy sequence converges, the original Cauchy sequence converges to the same limit.

Lemma 12.4 (Reduction to rapidly Cauchy sequences). Suppose that $C > 1$ and that every sequence $(g_k)_{k\in\mathbb{N}}$ in $L^p(X,\mu)$ satisfying

$$\|g_{k+1} - g_k\|_p^p \le C^{-k} \qquad \text{for all } k \in \mathbb{N} \tag{12.5}$$

converges in L^p to an element of $L^p(X,\mu)$. Then every Cauchy sequence in $L^p(X,\mu)$ converges in L^p, so $L^p(X,\mu)$ is complete.

Proof of Lemma 12.4. Suppose that $(f_n)_{n\in\mathbb{N}}$ is a Cauchy sequence in $L^p(X,\mu)$.

Step 1: Extraction of a rapidly Cauchy subsequence.
For each $k \in \mathbb{N}$, applying the Cauchy property of (f_n) with $\varepsilon = C^{-k}$ provides an index $N_k \in \mathbb{N}$ such that $\|f_n - f_m\|_p^p < C^{-k}$ for all $n, m \ge N_k$. We choose a strictly increasing sequence of indices $n_1 < n_2 < n_3 < \cdots$ such that $n_k \ge N_k$ for every $k \in \mathbb{N}$. Because both n_k and n_{k+1} are at least N_k, the subsequence $g_k := f_{n_k}$ satisfies

$$\|g_{k+1} - g_k\|_p^p = \|f_{n_{k+1}} - f_{n_k}\|_p^p < C^{-k} \qquad \text{for all } k \in \mathbb{N}.$$

By hypothesis, the sequence $(g_k)_{k\in\mathbb{N}}$ converges in L^p to an element $f \in L^p(X,\mu)$.

Step 2: Convergence of the full sequence.
For an arbitrary $\varepsilon > 0$, we choose an index $N \in \mathbb{N}$ such that $\|f_n - f_m\|_p^p < \varepsilon$ for all $n, m \ge N$. Because $n_k \to \infty$ and $\|f_{n_k} - f\|_p^p \to 0$ as $k \to \infty$, there exists an index k such that $n_k \ge N$ and $\|f_{n_k} - f\|_p^p < \varepsilon$. For every $n \ge N$, applying the quasi-triangle inequality (12.3) gives

$$\|f_n - f\|_p^p \le K_p(\|f_n - f_{n_k}\|_p^p + \|f_{n_k} - f\|_p^p) < K_p(\varepsilon + \varepsilon) = 2K_p\varepsilon.$$

Because $\varepsilon > 0$ was arbitrary and K_p is a fixed positive constant, it follows that $\|f_n - f\|_p^p \to 0$ as $n \to \infty$. This proves that $f_n \to f$ in $L^p(X,\mu)$. □

12.2 The Riesz–Fischer theorem

Theorem 12.5 (Riesz–Fischer theorem). For every $p \in (0, +\infty)$, the space $L^p(X,\mu)$ is complete: every Cauchy sequence in $L^p(X,\mu)$ converges in L^p to an element of $L^p(X,\mu)$.

By Lemma 12.4, it suffices to prove that every sequence satisfying (12.5) converges in $L^p(X,\mu)$ for one convenient choice of $C > 1$. We set $C := 2K_p$. With this choice, the geometric factor K_p^n from Lemma 12.3

is overwhelmed by the geometric decay C^{-n}: the product satisfies $K_p^n C^{-n} = 2^{-n}$, which forms a summable sequence. We then reconstruct the sequence via the telescoping identity

$$f_n(x) = f_1(x) + \sum_{k=1}^{n-1} (f_{k+1}(x) - f_k(x))$$

at almost every point $x \in X$.

Proof of Theorem 12.5. We define $C \coloneqq 2K_p > 1$. Suppose that $(f_n)_{n\in\mathbb{N}}$ is a sequence in $L^p(X,\mu)$ satisfying

$$\|f_{n+1} - f_n\|_p^p \le C^{-n} \qquad \text{for all } n \in \mathbb{N}.$$

We fix a representative function for each f_n, which is a measurable function $f_n \colon X \to \mathbb{R}$, and define the non-negative measurable functions $h_n \coloneqq |f_{n+1} - f_n| \colon X \to [0, +\infty)$.

Step 1: Almost-everywhere convergence of the series of absolute differences.
We define the extended real-valued function $G \colon X \to [0, +\infty]$ by $G(x) \coloneqq \sum_{n=1}^{+\infty} h_n(x)$. For every $N \in \mathbb{N}$, applying the iterated quasi-triangle inequality (Lemma 12.3) pointwise to $h_1(x), \dots, h_N(x)$ and integrating over X yields

$$\int_X \left(\sum_{n=1}^{N} h_n \right)^p \mathrm{d}\mu \le \sum_{n=1}^{N} K_p^n \int_X h_n^p \, \mathrm{d}\mu \le \sum_{n=1}^{N} K_p^n C^{-n} = \sum_{n=1}^{N} 2^{-n} \le 1.$$

As $N \to \infty$, the partial sums $\sum_{n=1}^{N} h_n$ increase pointwise to G, and their p-th powers increase pointwise to G^p. Applying the monotone convergence theorem gives $\int_X G^p \, \mathrm{d}\mu \le 1$. Because G^p is a non-negative function with finite integral, it is finite almost everywhere (Proposition 21.1 / Note 21). Consequently, the set

$$E \coloneqq \{x \in X : G(x) = +\infty\}$$

satisfies $\mu(E) = 0$.

Step 2: Pointwise definition of the candidate limit.
For every point $x \in X \setminus E$, the series of real numbers $\sum_{k=1}^{+\infty} (f_{k+1}(x) - f_k(x))$ is absolutely convergent because $\sum_{k=1}^{\infty} h_k(x) = G(x) < +\infty$. Because $\mathbb{R}$ is complete, the partial sums

$$f_n(x) = f_1(x) + \sum_{k=1}^{n-1} (f_{k+1}(x) - f_k(x))$$

converge to a real number as $n \to \infty$. We define the function $f \colon X \to \mathbb{R}$ by

$$f(x) \coloneqq \begin{cases} \lim_{n\to\infty} f_n(x), & x \in X \setminus E, \\ 0, & x \in E. \end{cases}$$

The function f is measurable as the pointwise limit of the sequence of measurable functions $f_n \mathbb{1}_{X\setminus E}$, and $f_n \to f$ pointwise almost everywhere on X.

Step 3: Tail estimates and convergence in L^p.
For each $n \in \mathbb{N}$, we define the tail function $T_n \colon X \to [0, +\infty]$ by $T_n(x) \coloneqq \sum_{k=n}^{+\infty} h_k(x)$. For every $x \in X \setminus E$ and $m > n$, the triangle inequality gives

$$|f_m(x) - f_n(x)| \le \sum_{k=n}^{m-1} h_k(x) \le T_n(x).$$

Passing to the limit as $m \to \infty$ yields

$$|f(x) - f_n(x)| \le T_n(x) \qquad \text{for all } x \in X \setminus E. \tag{12.6}$$

For every integer $M \in \mathbb{N}$, applying Lemma 12.3 to the $M+1$ non-negative numbers $h_n(x), h_{n+1}(x), \dots, h_{n+M}(x)$ assigns the weight K_p^{j+1} to $h_{n+j}(x)$. Integrating over X yields

$$\int_X \left(\sum_{j=0}^{M} h_{n+j} \right)^p \mathrm{d}\mu \leq \sum_{j=0}^{M} K_p^{j+1} \int_X h_{n+j}^p \, \mathrm{d}\mu \leq \sum_{j=0}^{M} K_p^{j+1} C^{-(n+j)} = K_p C^{-n} \sum_{j=0}^{M} 2^{-j} \leq 2K_p C^{-n}.$$

As $M \to \infty$, the partial sums increase pointwise to T_n, so the monotone convergence theorem gives $\int_X T_n^p \, \mathrm{d}\mu \leq 2K_p C^{-n}$. Because $\mu(E) = 0$, the bound (12.6) holds almost everywhere on X, and therefore

$$\int_X |f - f_n|^p \, \mathrm{d}\mu \leq \int_X T_n^p \, \mathrm{d}\mu \leq 2K_p C^{-n}. \tag{12.7}$$

Step 4: Membership in L^p and completion of the proof.
By (12.7), the function $f - f_n$ belongs to $\mathcal{L}^p(X, \mu)$. Because $\mathcal{L}^p(X, \mu)$ is a vector space (Note 24), the sum $f = f_n + (f - f_n)$ belongs to $\mathcal{L}^p(X, \mu)$. Since $C > 1$, the estimate (12.7) gives

$$\|f_n - f\|_p^p \leq 2K_p C^{-n} \longrightarrow 0 \qquad \text{as } n \to \infty,$$

which proves that $f_n \to f$ in $L^p(X, \mu)$. Applying Lemma 12.4 concludes the proof that every Cauchy sequence in $L^p(X, \mu)$ converges in $L^p(X, \mu)$. □

Corollary 12.6 (L^p as a complete space). For $p \in [1, +\infty)$, the normed space $(L^p(X, \mu), \|\cdot\|_p)$ is complete; a complete normed space is called a **Banach space**. For $p \in (0, 1)$, the space $L^p(X, \mu)$ equipped with the metric $d_p(f, g) = \|f - g\|_p^p$ is a complete metric space.

In particular, $L^1([a, b], \mathcal{L})$ is a Banach space. This completeness resolves the failure of completeness of $C^0([a, b])$ under the L^1 metric observed in Note 1: every Cauchy sequence of continuous functions converges in $L^1([a, b], \mathcal{L})$ to an element of $L^1([a, b], \mathcal{L})$, even when that limit cannot be represented by any continuous function.

The proof of Theorem 12.5 established more than convergence in the L^p metric. In Step 2, the rapidly Cauchy sequence (f_n) was shown to converge to f pointwise almost everywhere. Combined with the subsequence extraction in Lemma 12.4, this proves that every Cauchy sequence in $L^p(X, \mu)$ possesses a subsequence that converges almost everywhere to its L^p limit. The next section examines this relationship in detail.

12.3 L^p convergence and pointwise convergence

On a measure space $(X, \mathcal{A}, \mu)$, a sequence of functions $(f_n)_{n \in \mathbb{N}}$ can approach a limit function f in two distinct topological senses:

1. **Pointwise convergence almost everywhere:** the numerical sequence $f_n(x)$ converges to $f(x)$ for all points $x \in X$ outside a set of measure zero.
2. **Convergence in L^p:** the integrated average deviation $\int_X |f_n - f|^p \, \mathrm{d}\mu$ tends to 0 as $n \to \infty$.

Convergence in L^p allows $f_n(x)$ to deviate substantially from $f(x)$ on sets of small measure, provided the total integral remains small. In this section, we determine the precise connection between these two modes of convergence.

12.3.1 Rapid L^p convergence implies almost-everywhere convergence

While convergence in L^p alone does not imply pointwise convergence everywhere, convergence with summable errors ensures convergence almost everywhere.

Theorem 12.7 (Rapid L^p convergence implies almost-everywhere convergence). Suppose that $f_n, f \in L^p(X,\mu)$ satisfy

$$\sum_{n=1}^{+\infty} \|f_n - f\|_p^p < +\infty. \tag{12.8}$$

Then $f_n \to f$ almost everywhere as $n \to \infty$.

The proof converts summability of the integrals $\int_X |f_n - f|^p \, \mathrm{d}\mu$ into summability of the pointwise numerical terms $|f_n(x) - f(x)|^p$ outside a null set; the terms of any convergent numerical series necessarily tend to 0.

Proof of Theorem 12.7. We define the extended real-valued function $S\colon X \to [0, +\infty]$ by

$$S(x) \coloneqq \sum_{n=1}^{+\infty} |f_n(x) - f(x)|^p.$$

The partial sums of this series form a non-decreasing sequence of non-negative measurable functions that increase pointwise to S. Applying the monotone convergence theorem and the additivity of the integral for non-negative functions yields

$$\int_X S \, \mathrm{d}\mu = \sum_{n=1}^{+\infty} \int_X |f_n - f|^p \, \mathrm{d}\mu = \sum_{n=1}^{+\infty} \|f_n - f\|_p^p < +\infty.$$

Because S is non-negative and has finite integral, it is finite almost everywhere (Proposition 21.1 / Note 21). Thus there exists a measurable set $N \subseteq X$ with $\mu(N) = 0$ such that $S(x) < +\infty$ for all $x \in X \setminus N$. At every point $x \in X \setminus N$, the convergence of the numerical series $\sum_{n=1}^{\infty} |f_n(x) - f(x)|^p$ implies that its general term tends to zero: $|f_n(x) - f(x)|^p \to 0$. Because $t \mapsto t^{1/p}$ is continuous at 0, it follows that $f_n(x) \to f(x)$ for every $x \in X \setminus N$, which proves that $f_n \to f$ almost everywhere. □

Corollary 12.8 (Almost-everywhere convergent subsequences). If $f_n \to f$ in $L^p(X,\mu)$, then there exists a subsequence $(f_{n_k})_{k\in\mathbb{N}}$ such that $f_{n_k} \to f$ almost everywhere as $k \to \infty$.

Proof of Corollary 12.8. Because $\|f_n - f\|_p^p \to 0$ as $n \to \infty$, for each $k \in \mathbb{N}$ we may select an index n_k such that $n_1 < n_2 < n_3 < \cdots$ and $\|f_{n_k} - f\|_p^p < 2^{-k}$. The resulting subsequence satisfies

$$\sum_{k=1}^{+\infty} \|f_{n_k} - f\|_p^p < \sum_{k=1}^{+\infty} 2^{-k} = 1 < +\infty.$$

Applying Theorem 12.7 to this subsequence implies that $f_{n_k} \to f$ almost everywhere. □

Corollary 12.9 (Agreement of L^p and almost-everywhere limits). If a sequence $(f_n)_{n\in\mathbb{N}}$ satisfies $f_n \to f$ in $L^p(X,\mu)$ and $f_n \to g$ almost everywhere, then $f = g$ almost everywhere.

Proof of Corollary 12.9. By Corollary 12.8, there exists a subsequence $(f_{n_k})_{k\in\mathbb{N}}$ that converges to f outside a null set $N_1 \subseteq X$. Because $f_n \to g$ almost everywhere, the full sequence (f_n) converges to g outside a null set $N_2 \subseteq X$. The subsequence (f_{n_k}) therefore converges to g on $X \setminus N_2$. For every point

$x \in X \setminus (N_1 \cup N_2)$, the sequence of real numbers $(f_{n_k}(x))$ has both limits $f(x)$ and $g(x)$. By uniqueness of limits in $\mathbb{R}$, $f(x) = g(x)$. Because $\mu(N_1 \cup N_2) \leq \mu(N_1) + \mu(N_2) = 0$, this proves that $f = g$ almost everywhere. □

Corollary 12.9 guarantees that the two modes of convergence never yield incompatible limits: whenever a sequence converges both in $L^p(X, \mu)$ and pointwise almost everywhere, the two limits represent the same element of $L^p(X, \mu)$.

12.3.2 L^p convergence without pointwise convergence

Corollary 12.8 establishes almost-everywhere convergence only along a subsequence. The following classical construction demonstrates that passing to a subsequence is strictly necessary: an L^p-convergent sequence may fail to converge at every point of the domain.

Example 12.10 (The typewriter sequence). Consider the interval $[0, 1)$ equipped with Lebesgue measure $\mathcal{L}$. Every positive integer $n \in \mathbb{N}$ can be written uniquely in the form $n = 2^k + j$ with integers $k \geq 0$ and $0 \leq j < 2^k$. We define the sequence of indicator functions

$$f_n \coloneqq \mathbb{1}_{I_n}, \qquad I_n \coloneqq \left[\frac{j}{2^k}, \frac{j+1}{2^k}\right), \qquad n = 2^k + j.$$

For a fixed integer $k \geq 0$, the indices $n = 2^k, 2^k + 1, \ldots, 2^{k+1} - 1$ constitute the k-th *generation*. The intervals I_n of generation k have length 2^{-k} and partition $[0, 1)$ from left to right, as illustrated in Figure 12.1.

Convergence in L^p. For $n = 2^k + j$, the L^p norm evaluates to

$$\|f_n\|_p^p = \int_{[0,1)} \mathbb{1}_{I_n} \, \mathrm{d}\mathcal{L} = \mathcal{L}(I_n) = 2^{-k}.$$

Because $n < 2^{k+1}$, we have $2^{-k} < 2/n$. Therefore $\|f_n - 0\|_p^p < 2/n \to 0$ as $n \to \infty$, which proves that $f_n \to 0$ in $L^p([0, 1), \mathcal{L})$.

Divergence at every point. Let $x \in [0, 1)$ be an arbitrary point. In each generation k, the intervals I_n partition $[0, 1)$, so x belongs to exactly one interval of that generation. Consequently, $f_n(x) = 1$ for infinitely many indices n. On the other hand, for each generation $k \geq 1$, there are $2^k - 1 \geq 1$ intervals that do not contain x, so $f_n(x) = 0$ for infinitely many indices n. Because the sequence $(f_n(x))_{n \in \mathbb{N}}$ alternates between 0 and 1 infinitely often, it does not converge. This divergence holds at every point $x \in [0, 1)$.

A rapidly convergent subsequence. Choosing the first interval of each generation, $n_k \coloneqq 2^k$, gives the subsequence $f_{n_k} = \mathbb{1}_{[0, 2^{-k})}$. This subsequence satisfies

$$\sum_{k=0}^{+\infty} \|f_{n_k}\|_p^p = \sum_{k=0}^{+\infty} 2^{-k} = 2 < +\infty.$$

In accordance with Theorem 12.7, $f_{n_k}(x) \to 0$ for every $x \in (0, 1)$, that is, almost everywhere on $[0, 1)$. For the full sequence, the sum of errors diverges: $\sum_{n=1}^{\infty} \|f_n\|_p^p = \sum_{k \geq 0} 2^k \cdot 2^{-k} = \sum_{k \geq 0} 1 = +\infty$, explaining why Theorem 12.7 does not apply to the whole sequence.

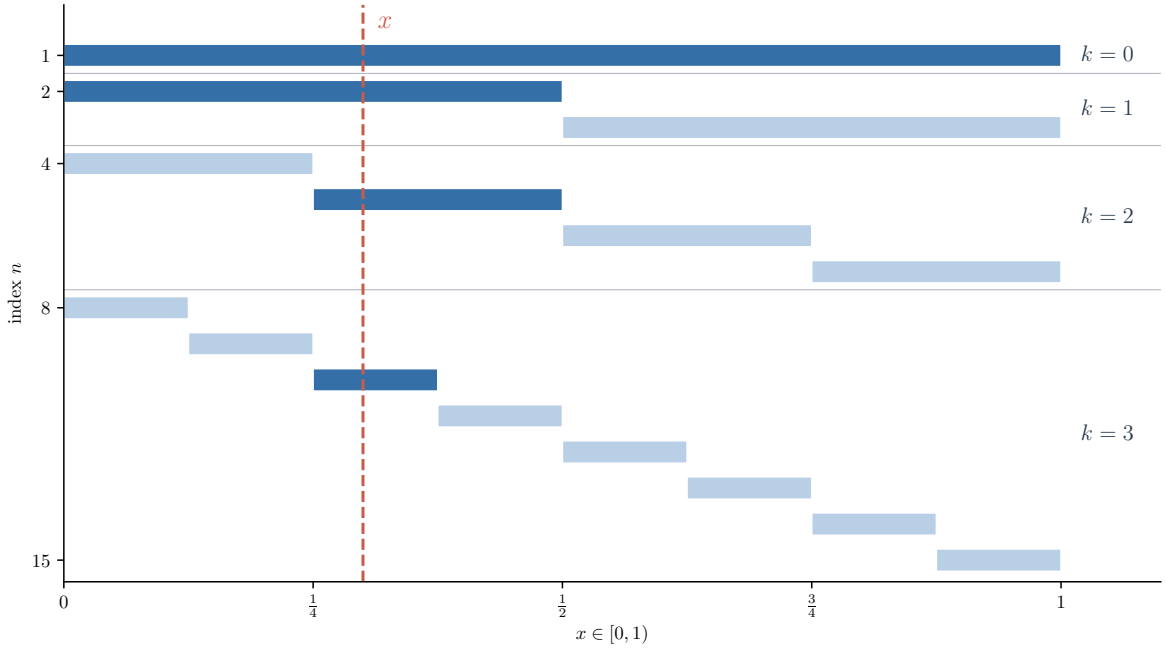


Figure 12.1: The intervals I_n of the typewriter sequence for $n = 1, \dots, 15$, grouped by generation k. Each generation sweeps across $[0, 1)$ with intervals of length 2^{-k}. A fixed point x (dashed line) lies in exactly one interval of every generation (dark bars), so $f_n(x) = 1$ infinitely often.

Figure description: A chart with the index n on the vertical axis and the interval from 0 to 1 on the horizontal axis. Each row n shows a horizontal bar over the interval I sub n. Rows are grouped into generations k equals 0 to 3, with one bar of full length, then two halves, four quarters, and eight eighths. A vertical dashed line at a fixed point x crosses one dark bar in every generation.

12.3.3 Pointwise convergence without L^p convergence

Conversely, a sequence of functions may converge pointwise almost everywhere to a limit function without converging in the L^p metric. Two distinct analytical mechanisms produce this behavior: the escape of mass to infinity on spaces of infinite measure, and the concentration of mass on shrinking sets.

Example 12.11 (Escape of mass and concentration of mass in L^p). We exhibit both mechanisms through concrete sequences on subsets of the real line.

Escape of mass to infinity. On $\mathbb{R}$ equipped with Lebesgue measure, consider the translated indicator functions

$$f_n \coloneqq \mathbb{1}_{[n,n+1)}, \qquad n \in \mathbb{N}.$$

For each fixed $x \in \mathbb{R}$, we have $x < n$ for all sufficiently large n, so $x \notin [n, n+1)$ and $f_n(x) = 0$. Thus $f_n(x) \to 0$ for every $x \in \mathbb{R}$. However, the L^p norm remains constant:

$$\|f_n\|_p^p = \int_{\mathbb{R}} \mathbb{1}_{[n,n+1)} \, \mathrm{d}\mathcal{L} = \mathcal{L}([n, n+1)) = 1 \qquad \text{for all } n \in \mathbb{N}.$$

Concentration of mass on a finite measure space. On the bounded interval $(0, 1]$ equipped with Lebesgue measure, consider the functions

$$g_n \coloneqq 2^{n/p} \, \mathbb{1}_{[2^{-n}, 2^{1-n})}, \qquad n \in \mathbb{N}.$$

For every point $x \in (0, 1]$, there exists a unique integer $n_0 \in \mathbb{N}$ such that $x \in [2^{-n_0}, 2^{1-n_0})$. For all $n > n_0$, we have $x \geq 2^{1-n_0} > 2^{1-n}$, so $g_n(x) = 0$. Thus $g_n(x) \to 0$ pointwise everywhere on $(0, 1]$. However, the

p-th power integral scales to compensate for the shrinking interval:

$$\|g_n\|_p^p = \int_{(0,1]} 2^n \, \mathbb{1}_{[2^{-n},2^{1-n})} \, \mathrm{d}\mathcal{L} = 2^n \cdot \mathcal{L}([2^{-n}, 2^{1-n})) = 2^n \cdot 2^{-n} = 1 \qquad \text{for all } n \in \mathbb{N}.$$

Neither sequence converges in L^p to any limit. Indeed, suppose that $f_n \to h$ in $L^p(\mathbb{R})$ for some function h. Because $f_n \to 0$ pointwise everywhere, Corollary 12.9 implies that $h = 0$ almost everywhere. Then $\|f_n\|_p^p = \|f_n - h\|_p^p \to 0$, which contradicts $\|f_n\|_p^p = 1$ for all n. An identical argument shows that (g_n) cannot converge in $L^p((0,1])$. By Theorem 12.5, neither sequence is a Cauchy sequence in L^p.

The sequence (g_n) in Example 12.11 demonstrates that even on a space of finite measure, pointwise convergence almost everywhere does not imply convergence in L^p.

12.3.4 Domination restores L^p convergence

In our study of integral limit theorems (Note 22), the dominated convergence theorem prevented both escape of mass to infinity and mass concentration by requiring a single dominating integrable function. Applying that theorem to the power difference $|f_n - f|^p$ establishes that almost-everywhere convergence implies L^p convergence whenever the sequence is dominated in L^p.

Proposition 12.12 (Dominated convergence in L^p). Suppose that measurable functions $f_n \colon X \to \mathbb{R}$ converge almost everywhere to a measurable function $f \colon X \to \mathbb{R}$, and that there exists a dominating function $g \in L^p(X, \mu)$ such that

$$|f_n(x)| \le g(x) \qquad \text{for almost every } x \in X \text{ and every } n \in \mathbb{N}.$$

Then $f_n, f \in L^p(X, \mu)$ and $f_n \to f$ in $L^p(X, \mu)$.

Proof of Proposition 12.12. Because $|f_n|^p \le g^p$ almost everywhere and $\int_X g^p \, \mathrm{d}\mu < +\infty$, each f_n belongs to $\mathcal{L}^p(X, \mu)$. Passing to the limit as $n \to \infty$ in $|f_n(x)| \le g(x)$ gives $|f(x)| \le g(x)$ almost everywhere, so $f \in \mathcal{L}^p(X, \mu)$. We define the sequence of non-negative measurable functions $u_n \colon X \to [0, +\infty)$ by $u_n(x) \coloneqq |f_n(x) - f(x)|^p$. Because $f_n \to f$ almost everywhere, $u_n \to 0$ almost everywhere on X. Furthermore, the triangle inequality in $\mathbb{R}$ yields

$$u_n(x) = |f_n(x) - f(x)|^p \le \big(|f_n(x)| + |f(x)|\big)^p \le \big(2g(x)\big)^p = 2^p g(x)^p \qquad \text{almost everywhere.}$$

Because $g \in L^p(X, \mu)$, the function $2^p g^p$ is integrable on X: $\int_X 2^p g^p \, \mathrm{d}\mu = 2^p \|g\|_p^p < +\infty$. Applying the dominated convergence theorem for L^1 functions (Theorem 22.1 / Note 22) to the sequence $(u_n)_{n \in \mathbb{N}}$ with dominating function $2^p g^p$ yields

$$\lim_{n \to \infty} \|f_n - f\|_p^p = \lim_{n \to \infty} \int_X u_n \, \mathrm{d}\mu = \int_X \lim_{n \to \infty} u_n \, \mathrm{d}\mu = 0.$$

This proves that $f_n \to f$ in $L^p(X, \mu)$. □

In Example 12.11, the minimal candidate dominating functions $\sup_n |f_n| = \mathbb{1}_{[1,+\infty)}$ and $\sup_n |g_n|$ fail to belong to L^p: the former has infinite measure support on $\mathbb{R}$, and the p-th power of the latter produces the non-integrable singularity $x \mapsto 1/x$ near 0. On a finite measure space, any sequence that is uniformly bounded by a constant M is dominated by the constant function $g \equiv M$, which belongs to $L^p(X, \mu)$ because $\int_X M^p \, \mathrm{d}\mu = M^p \mu(X) < +\infty$.

Remark (Summary of relations between L^p and almost-everywhere convergence). For $p \in (0, +\infty)$ and functions $f_n, f \in L^p(X, \mu)$:

- **Rapid L^p convergence implies almost-everywhere convergence:** If $\sum_{n=1}^{\infty} \|f_n - f\|_p^p < +\infty$, then $f_n \to f$ almost everywhere (Theorem 12.7).
- **Subsequential convergence:** If $f_n \to f$ in L^p, then a subsequence converges to f almost everywhere (Corollary 12.8), although the full sequence may converge at no point (Example 12.10).
- **Pointwise convergence does not imply L^p convergence:** Almost-everywhere convergence $f_n \to f$ does not guarantee L^p convergence, even on a space of finite measure (Example 12.11); however, L^p convergence is restored when the sequence is dominated by a function in L^p (Proposition 12.12).
- **Consistency of limits:** Whenever a sequence converges both in L^p and almost everywhere, the two limits coincide almost everywhere (Corollary 12.9).